



# **Engineering Mathematics**



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# Matrices and Determinants

## 1.1 BASIC CONCEPTS OF MATRICES

A Matrix is a rectangular array of numbers or elements, i.e., arrangement of numbers or elements in rows and columns.

Matrix having  $m$  rows and  $n$  columns is of order  $m \times n$  and total number of elements in matrix are  $mn$ . (Read as  $m$  cross  $n$  or  $m$  by  $n$ ). A matrix is denoted by capital letter. The elements of the matrix are enclosed within the square brackets [ ].

$$\text{Ex : } A = \begin{bmatrix} 2 & 5 & -4 \\ 1 & -8 & 3 \end{bmatrix} \begin{matrix} \rightarrow 1^{\text{st}} \text{ row} \\ \rightarrow 2^{\text{nd}} \text{ row} \end{matrix}$$

$$\begin{matrix} \downarrow & \downarrow & \downarrow \\ 1^{\text{st}} \text{ col} & 2^{\text{nd}} \text{ col} & 3^{\text{rd}} \text{ col} \end{matrix}$$

It has 2 rows and 3 columns. Therefore the order of matrix  $A$  is  $2 \times 3$ .

## 1.2 TYPES OF MATRIX

**1. Row Matrix :** A matrix having only one row is called a row matrix. Ex.  $A = [8 \ -1 \ 5]$  is a row matrix of order  $1 \times 3$ .

**2. Column Matrix :** A matrix having only one column

$$\text{is called a column matrix. Ex. } A = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} \text{ is a column}$$

matrix of order  $3 \times 1$ .

**3. Square Matrix :** A matrix having equal numbers of rows and columns is a square matrix.

$$\text{Ex : } A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}_{2 \times 2} \quad B = \begin{bmatrix} 1 & 5 & 8 \\ 3 & -2 & 1 \\ 5 & 8 & 7 \end{bmatrix}_{3 \times 3}$$

**4. Diagonal Matrix :** A square matrix whose all the elements except the diagonal elements are zero is called a diagonal matrix.

$$\text{Ex : } A = \begin{bmatrix} 5 & 0 \\ 0 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

**5. Scalar Matrix :** It is a diagonal matrix in which all the elements in the principal diagonal are same.

$$\text{Ex : } A = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix} \text{ is a scalar matrix of order } 3 \times 3$$

**6. Unit Matrix or Identity Matrix :** It is denoted by  $I$ . It is a square matrix whose all non diagonal elements are zero and diagonal elements are 1 each.

$$\text{Ex : } I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2} \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

**7. Zero Matrix or Null Matrix :** A matrix whose all the elements are zero is known as a zero matrix. It is denoted by  $0$ .

$$\text{Ex : } 0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

**8. Equal Matrices :** The matrices  $A$  and  $B$  are said to be equal if their orders are same and elements at the corresponding places are equal.

$$\text{Ex : if } A = \begin{bmatrix} 3 & 4 & 5 \\ 8 & 1 & 2 \end{bmatrix}_{2 \times 3} \text{ and } B = \begin{bmatrix} 3 & 4 & 5 \\ 8 & 1 & 2 \end{bmatrix}_{2 \times 3} \\ \text{then } A = B$$

### 1.3 OPERATION ON MATRICES

**1. Scalar multiplication :** If  $k$  is a scalar (i.e., a single number) and  $A$  is a matrix of any size, then  $kA$  is a matrix obtained by multiplying each elements of  $A$  by the scalar  $k$ .

**Illustration (1)**

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ then,}$$

$$kA = k \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} ka_{11} & ka_{12} \\ ka_{21} & ka_{22} \end{bmatrix}$$

**Illustration (2)**

$$\text{If } A = \begin{bmatrix} 4 & 2 & 1 \\ 5 & 8 & -2 \end{bmatrix} \text{ then,}$$

$$3A = 3 \begin{bmatrix} 4 & 2 & 1 \\ 5 & 8 & -2 \end{bmatrix} = \begin{bmatrix} 12 & 6 & 3 \\ 15 & 24 & -6 \end{bmatrix}$$

**Illustration (3)**

$$\text{If } A = \begin{bmatrix} 4 & 2 \\ -1 & 3 \end{bmatrix} \text{ then,}$$

$$\frac{1}{2}A = \frac{1}{2} \begin{bmatrix} 4 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} \frac{4}{2} & \frac{2}{2} \\ \frac{-1}{2} & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix}$$

**2. Addition of matrices :** If  $A$  and  $B$  are two matrices of the same order, then their sum  $A + B$  is defined and is obtained by adding the corresponding elements of  $A$  and  $B$ .

$$\text{Illustration, If } A = \begin{bmatrix} 3 & 2 & 1 \\ 5 & 1 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 & 5 \\ 3 & 1 & 4 \end{bmatrix}$$

$$\begin{aligned} \text{then, } A + B &= \begin{bmatrix} 3 & 2 & 1 \\ 5 & 1 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 5 \\ 3 & 1 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 3+2 & 2+0 & 1+5 \\ 5+3 & 1+1 & 4+4 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 6 \\ 8 & 2 & 8 \end{bmatrix} \end{aligned}$$

**3. Subtraction of Matrices :** If  $A$  and  $B$  are two matrices of the same order, then their difference  $A - B$  is defined and is obtained by subtracting the elements of  $B$  from the corresponding elements of  $A$ .

$$\text{Illustration, If } A = \begin{bmatrix} 2 & 1 \\ 3 & 8 \\ 9 & 12 \end{bmatrix} \text{ and } B = \begin{bmatrix} 6 & 1 \\ 5 & 3 \\ -1 & 4 \end{bmatrix}$$

$$\text{then, } A - B = \begin{bmatrix} 2 & 1 \\ 3 & 8 \\ 9 & 12 \end{bmatrix} - \begin{bmatrix} 6 & 1 \\ 5 & 3 \\ -1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2-6 & 1-1 \\ 3-5 & 8-3 \\ 9-(-1) & 12-4 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ -2 & 5 \\ 10 & 8 \end{bmatrix}$$

### Properties of addition of Matrices

If  $A, B, C$  and  $D$  are matrices of same order. Where,  $O$  is the null matrix, then

i) **Commutative law of addition**

$$A + B = B + A$$

ii) **Associative law of addition**

$$(A + B) + C = A + (B + C)$$

iii) **Existence law of identity**

$$A + O = O + A = A$$

**4. Transpose of a matrix :** Let  $A$  be a matrix of order  $m \times n$ . The  $n \times m$  matrix obtained from  $A$  by interchanging its rows and columns is called the **transpose of matrix** and is denoted by  $A'$  or by  $A^T$ .

$$\text{Illustration, let } A = \begin{bmatrix} 2 & -1 \\ 3 & 1 \\ 4 & 5 \end{bmatrix}_{(3 \times 2)}$$

The transpose of  $A$  is  $2 \times 3$  matrix

$$A^T = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 1 & 5 \end{bmatrix}_{(2 \times 3)}$$



$$\therefore 4 - \lambda = 0 \quad \text{and} \quad 3 - \lambda = 0$$

$$\lambda = 4 \quad \lambda = +3$$

**60. Find the characteristic equation and its roots**

**of a square matrix**  $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ .

**Solution :** Given  $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

The characteristic equation of  $A$  is,  $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(1 - \lambda) - (2 \times 2) = 0$$

$$1 - \lambda - \lambda + \lambda^2 - 4 = 0$$

$\lambda^2 - 2\lambda - 3 = 0$  is characteristic equation.

$$\lambda^2 + 1\lambda - 3\lambda - 3 = 0 \text{ (by factorization)}$$

$$\lambda(\lambda + 1) - 3(\lambda + 1) = 0$$

$$(\lambda + 1)(\lambda - 3) = 0$$

$$\lambda + 1 = 0 \text{ or } \lambda - 3 = 0$$

$$\lambda = -1 \text{ or } \lambda = 3$$

$\therefore \lambda = -1, 3$  are characteristic roots of matrix  $A$ .

### PRACTICE MCQ's

**1. If**  $A = \begin{bmatrix} -1 & 7 \\ -4 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$  **then**  $A + B$  **is**

a)  $\begin{bmatrix} 1 & 10 \\ 1 & 5 \end{bmatrix}$       b)  $\begin{bmatrix} 1 & 1 \\ 10 & 5 \end{bmatrix}$

c)  $\begin{bmatrix} -1 & -1 \\ 10 & 5 \end{bmatrix}$       d)  $\begin{bmatrix} -1 & 10 \\ -1 & 5 \end{bmatrix}$

**2. If**  $A = \begin{bmatrix} 4 & -3 \\ 2 & 1 \end{bmatrix}$  **and**  $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ , **then**  $2A + B$  **is**

a)  $\begin{bmatrix} 9 & -4 \\ 4 & 3 \end{bmatrix}$       b)  $\begin{bmatrix} 3 & -1 \\ 5 & 7 \end{bmatrix}$

c)  $\begin{bmatrix} 9 & 4 \\ -4 & 3 \end{bmatrix}$       d)  $\begin{bmatrix} 3 & -4 \\ -4 & 9 \end{bmatrix}$

**3. If**  $A = \begin{bmatrix} 4 & -6 \\ -2 & 7 \end{bmatrix}$  **then**  $A + A'$  **is**

a)  $\begin{bmatrix} 8 & -8 \\ -8 & 14 \end{bmatrix}$       b)  $\begin{bmatrix} 14 & -8 \\ -8 & -8 \end{bmatrix}$

c)  $\begin{bmatrix} -8 & 8 \\ 8 & 14 \end{bmatrix}$       d)  $\begin{bmatrix} 8 & 18 \\ 8 & -4 \end{bmatrix}$

**4. If**  $A = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ ,  $B = [3 \ 4]$ , **then**  $AB$  **is**

a)  $\begin{bmatrix} 3 & 3 \\ 12 & 16 \end{bmatrix}$       b)  $\begin{bmatrix} 3 & 4 \\ 12 & 16 \end{bmatrix}$

c)  $\begin{bmatrix} 4 & 3 \\ 16 & 12 \end{bmatrix}$       d)  $\begin{bmatrix} -3 & -4 \\ 16 & 12 \end{bmatrix}$

**5. The product of**  $A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & -1 & 3 \end{bmatrix}$  **and**  $B = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix}$

**is**

a)  $[10, 16]$       b)  $\begin{bmatrix} -10 \\ 16 \end{bmatrix}$

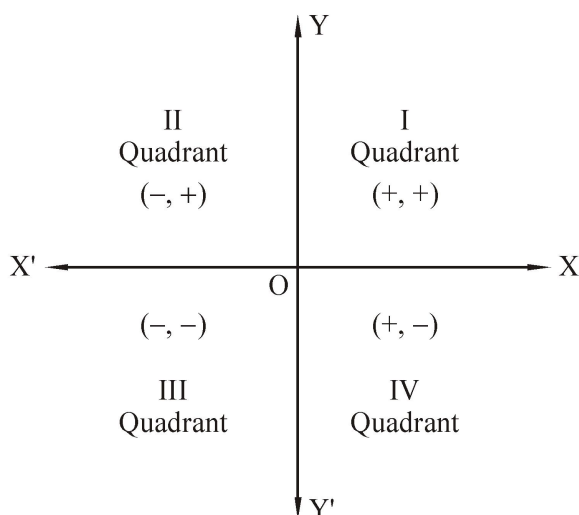
c)  $\begin{bmatrix} 10 \\ 16 \end{bmatrix}$       d)  $[16 \ 10]$

# Straight Lines

## 2.1 CO-ORDINATE SYSTEM

Analytical geometry deals with the study of the geometrical properties of points and lines by the use of algebraic methods. Thus, it is called algebraic geometry or co-ordinate geometry.

The position of an object or a point is located using cartesian plane



## 2.2 SOME IMPORTANT FORMULA

1. Distance of a point  $P(x, y)$  from origin  $O(0, 0)$

$$D = \sqrt{x^2 + y^2}$$

2. Distance between the points  $(x_1, y_1)$  and  $(x_2, y_2)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

3. Midpoint of the line,  $M = \left[ \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right]$

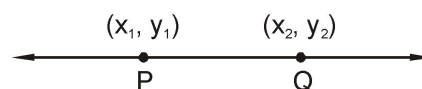
4. Centroid of the triangle

$$C = \left[ \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right]$$

## 2.3 STRAIGHT LINES

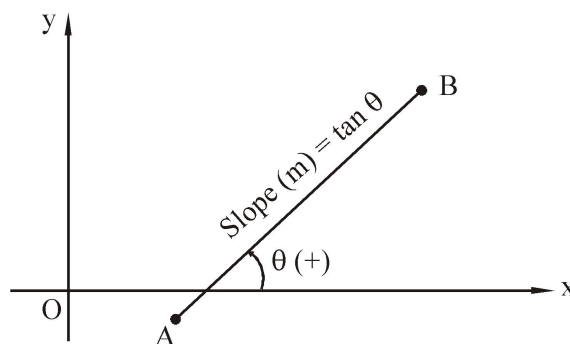
The straight line is the locus of point which moves without changing its direction

A straight line is a figure formed when two points  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  are connected between them, and the line ends are extended to infinity.



### Slope / Gradient of a straight line:

If a straight line makes an angle  $\theta$  with the positive direction of the x-axis, then  $\theta$  is called the inclination and tangent angle ( $\tan \theta$ ) is called the slope ( $m$ ) or gradient of the line.



The slope of a line is generally given as  $m = \tan \theta$

### Note :

1. Slope is **positive**, when  $\theta$  is an acute angle i.e.,  $\theta$  lies between  $0^\circ$  and  $90^\circ$ .
2. Slope is **negative**, when  $\theta$  is an obtuse angle i.e.,  $\theta$  lies between  $90^\circ$  and  $180^\circ$ .
3. Slope is **zero**, when  $\theta$  is equal to  $0^\circ$  or it is **parallel to x-axis**.
4. Slope is **infinite**, when  $\theta$  is equal to  $90^\circ$  or it is **perpendicular to x-axis**.

### EXAMPLE PROBLEMS

1. What is the slope of a line whose inclination is  $45^\circ$ .

**Solution :** Given  $\theta = 45^\circ$   
 $\therefore$  slope,  $m = \tan\theta = \tan 45^\circ$   
 $m = 1$

2. The slope of the line making an angle  $120^\circ$  with x-axis is

**Solution :**  
 Slope of the line,  $m = \tan\theta = \tan 120^\circ$   
 i.e.,  $m = \tan(180^\circ - 60^\circ) \Rightarrow m = -\tan 60^\circ$   
 $\therefore m = -\sqrt{3}$

3. The slope of the lines joining the points (3, 2) and (-4, 3) is

**Solution :**  
 $A(3, 2) = (x_1, y_1)$        $B(-4, 3) = (x_2, y_2)$   
 Slopes of line,  $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 2}{-4 - 3} = \frac{1}{-7} = -\frac{1}{7}$

4. Find the slope of the line joining the points (2, -3) (6, 1).

**Solution :** Given  $P(x_1, y_1) = (2, -3)$  and  $Q(x_2, y_2) = (6, 1)$

The slope of line,  $m = \frac{y_2 - y_1}{x_2 - x_1}$   
 $m = \frac{1 - (-3)}{6 - 2} = \frac{1 + 3}{4} = \frac{4}{4}$   
 $\therefore m = 1$

5. Find the slope of the line parallel to the line joining the points (2, 3) (4, -1).

**Solution :** Given  $A(x_1, y_1) = (2, 3)$  and  $B(x_2, y_2) = (4, -1)$

Slope of line,  $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - 3}{4 - 2} = \frac{-4}{2} = -2$

w.k.t slope of the parallel lines are equal.

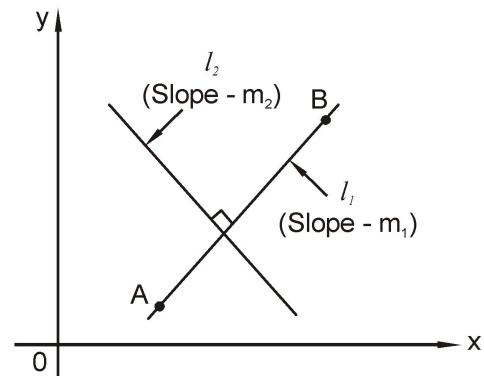
$$m_1 = m_2 = m = -2$$

$\therefore$  Slope of parallel line is -2

6. Find the slope of the line perpendicular to the line joining the points (-3, 6) and (5, -4).

**Solution :** Given  $A(x_1, y_1) = (-3, 6)$  and  $B(x_2, y_2) = (5, -4)$

$$\text{Slope of line, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 6}{5 + 3} = \frac{-10}{8} = -\frac{5}{4}$$



The slope of the perpendicular lines condition is

$$m_1 \times m_2 = -1$$

$$-\frac{5}{4} \times m_2 = -1$$

$$m_2 = \frac{-1(4)}{-5} = \frac{-4}{-5} = \frac{4}{5}$$

$\therefore$  Slope of perpendicular line is  $\frac{4}{5}$

7. If the slope of the line joining the points (2, 5) and (3,  $\lambda$ ) is -2, find the value of  $\lambda$ .

**Solution :** Given  $A(x_1, y_1) = (2, 5)$ ,  $B(x_2, y_2) = (3, \lambda)$  and  $m = -2$

$$\text{Slope of line, } m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow -2 = \frac{(\lambda - 5)}{(3 - 2)}$$

$$-2(3 - 2) = \lambda - 5$$

$$-6 + 4 = \lambda - 5$$

$$-2 + 5 = \lambda \quad \therefore \lambda = 3$$

8. Find the slope of the line  $3x - 4y + 8 = 0$ .

**Solution :** Given  $3x - 4y + 8 = 0$

# Trigonometry

Trigonometry is a branch of mathematics that deals with the study of the relationship between the sides and angle of a right angled triangle.

## 3.1 CONCEPT OF ANGLES AND MEASUREMENT

Angle can be defined as the rotation from the initial point to an endpoint of a ray.

Commonly used terms in describing the angles are:

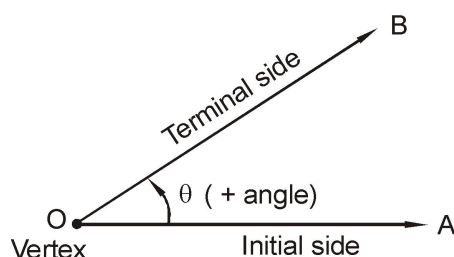
**Initial side :** The original ray

**Terminal side :** The final position of the ray after rotation

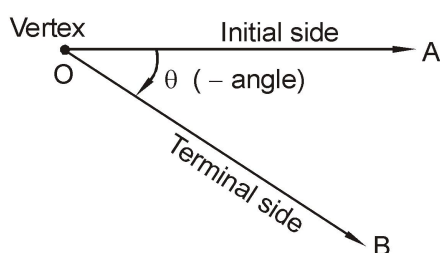
**Vertex :** Point of rotation

**Positive angle :** The direction of rotation is anticlockwise, then the angle is called positive angle.

**Negative angle :** The direction of rotation is clockwise, then the angle is called negative angle.



(i) Positive angle



(ii) Negative angle

The two commonly used units for measurement of angles are :

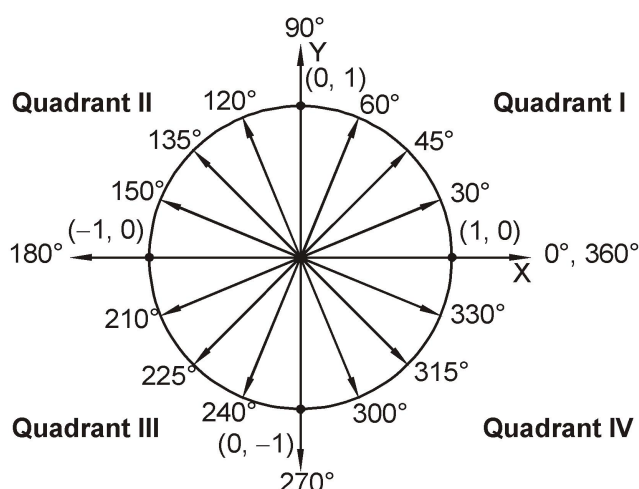
- i) Degree measurement
- ii) Radian measurement

### 3.1.1 Angle measurement - Degree Measure

A complete revolution, i.e., when the initial and terminal sides are in the same position after rotating clockwise, is divided into  $360^\circ$  units called degrees. So, if the rotation from the initial side to the terminal side is

$\left(\frac{1}{360}\right)^{\text{th}}$  of a revolution, the angle is said to be one degree.

It is denoted as  $1^\circ$



Time is measured in hours, minutes and seconds, where, 1 hour = 60 minutes and 1 minute = 60 seconds, similarly, while measuring angles,

- 1 degree = 60 minutes, denoted as  $1^\circ = 60'$
- 1 minute = 60 seconds, denoted as  $1' = 60''$

### 3.1.2 Angle measurement - Radian Measure

Radian is an angle subtended at the center of a circle by an arc, whose length of arc ( $s$ ) is equal to the radius ( $r$ ) of the circle.

### PREVIOUS YEAR MCQ's

1. If  $\tan x = 15/8$  and  $x$  is in the III quadrant then the value of  $(2\sin x - 3\cos x)/(2\cos x + 3\sin x)$  is (DCET - 2013)
  - a)  $61/6$                       b)  $-61/6$
  - c)  $-6/61$                     d)  $6/61$
2. The value of  $\{[\sin(2\pi - \theta) + \cos(-\theta)]/[\tan(-\theta) + \cot(2\pi + \theta)]\} - \{[\sin(\pi/2 + \theta) + \cos(3\pi/2 - \theta)]/[\cot(\pi + \theta) + \tan(2\pi - \theta)]\}$  is (DCET - 2013)
  - a) 0                              b) -1
  - c) +1                            d) -2
3. The value of  $\frac{(\sin 100^\circ + \sin 20^\circ)}{(\cos 100^\circ + \cos 20^\circ)}$  is (DCET - 2013)
  - a)  $\frac{\sqrt{3}}{2}$                           b)  $\frac{1}{2}$
  - c)  $\sqrt{3}$                             d) 1
4. If  $\sin A = 5/13$  and  $\sin B = 4/5$  then the value of  $\cos(A-B)$  is (DCET - 2013)
  - a)  $65/56$                       b)  $56/65$
  - c)  $16/65$                       d)  $-16/65$
5. On simplification the value of  $\frac{(\cos^3 A - \cos 3A)}{\cos A} + \frac{(\sin^3 A + \sin 3A)}{\sin A}$  is (DCET - 2013)
  - a) 3                                b) 1
  - c) 2                                d) 0
6. The value of  $\tan 225^\circ + \cot 135^\circ$  is (DCET-2014)
  - a) 1                                b) -1
  - c) 0                                d) 2
7. If  $\sin \theta = \frac{4}{5}$ ,  $\frac{\pi}{2} < \theta < \pi$ , then  $\frac{3\sin \theta + 4\cos \theta}{3\sin \theta - 4\cos \theta}$  is (DCET - 2014)
  - a)  $\infty$                       b)  $\frac{5}{24}$                       c)  $\frac{24}{5}$                       d) 0
8. The value of  $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$  is (DCET - 2014)
  - a) 0                                b)  $\cos 50^\circ$
  - c)  $\frac{1}{2}$                               d)  $\sin 50^\circ$
9. If  $\sin(45^\circ - \theta) = \frac{\sqrt{3} - 1}{2\sqrt{2}}$  and  $\theta$  is acute, then  $\theta$  is (DCET - 2014)
  - a)  $60^\circ$                       b)  $15^\circ$                       c)  $75^\circ$                       d)  $30^\circ$
10. If  $\tan \theta = \frac{\sin 2\theta}{1 + \cos 2\theta}$ , then  $\tan 22 \frac{1}{2}^\circ$  is (DCET - 2014)
  - a)  $\sqrt{2} - 1$                       b)  $\sqrt{2} + 1$
  - c)  $1 - \sqrt{2}$                       d)  $1 + \sqrt{3}$
11. The value of  $\sin 225^\circ + \cos(-135^\circ)$  is (DCET - 2015)
  - a)  $\sqrt{2}$                       b)  $-\sqrt{2}$                       c)  $\frac{1}{\sqrt{2}}$                       d)  $\frac{-1}{\sqrt{2}}$
12. The simplified value of  $\frac{\sin(180^\circ - A) \cot(90^\circ - A) \cos(360^\circ - A)}{\tan(180^\circ + A) \tan(90^\circ + A) \sin(-A)}$  is (DCET - 2015)
  - a)  $\sin A$                       b)  $-\sin A$
  - c) 1                                d)  $\operatorname{cosec} A$
13. The value of  $\frac{\sin 2A}{1 + \cos 2A}$  is (DCET - 2015)
  - a)  $2 \tan A$                       b)  $\sin A$
  - c)  $\cot A$                         d)  $\tan A$
14. If  $\tan A = \frac{3}{4}$  and  $\tan B = \frac{1}{7}$ , then the value of  $(A + B)$  is (DCET - 2015)
  - a)  $\frac{\pi}{6}$                       b)  $\frac{25}{23}$                       c)  $\frac{\pi}{4}$                       d)  $\frac{23}{25}$

# Integral Calculus and Applications

## 5.1 INDEFINITE INTEGRAL

Integration is the reverse process of differentiation.

Let  $y = f(x)$  be a function. If  $F(x)$  is the differential co-efficient of  $f(x)$  then  $f(x)$  itself is called as an integral, a primitive or anti-derivative of  $F(x)$  w.r.t. 'x'

It is denoted by  $\int F(x) dx = f(x)$

ie.,  $\int F(x) dx = f(x) \Leftrightarrow F(x) = f'(x)$

Since the derivative of a constant is zero

i.e.,  $\frac{d}{dx} (f(x) + c) = \frac{d}{dx} (f(x)) = F(x)$

$\Rightarrow \int F(x) dx = f(x) + C$

where 'C' is called a constant of integration.

In  $\int F(x) dx$ ,

$\int \rightarrow$  integral sign (elongated s)

$F(x) \rightarrow$  Integrand,  $dx \rightarrow$  w.r.t. 'x'

### Illustrations

$$1. \frac{d}{dx} (x) = 1 \Leftrightarrow \int 1 dx = x + c$$

$$2. \frac{d}{dx} (x^3) = 3x^2 \Leftrightarrow \int 3x^2 dx = x^3 + c$$

$$3. \frac{d}{dx} (\tan x) = \sec^2 x \Leftrightarrow \int \sec^2 x dx = \tan x + c$$

$$4. \frac{d}{dx} (\log x) = \frac{1}{x} \Leftrightarrow \int \frac{1}{x} dx = \log x + c$$

We already know the formulae for the derivatives of some standard functions. From these formulae, we can

write down the corresponding formulae for the integrals of these functions.

## 5.2 LIST OF STANDARD INTEGRALS

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

### Illustrations

$$(i) \int 1 dx = x + C$$

$$(ii) \int x dx = \frac{x^2}{2} + C$$

$$(iii) \int x^2 dx = \frac{x^3}{3} + C$$

$$(iv) \int \sqrt{x} dx = \frac{x^{3/2}}{3/2} + C$$

$$(v) \int x^{3/2} dx = \frac{x^{5/2}}{5/2} + C$$

$$2. \int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$$

$$3. \int \frac{1}{x} dx = \log x + C$$

$$4. \int \frac{1}{x^2} dx = -\frac{1}{x} + C$$

$$5. \int a^x dx = \frac{a^x}{\log a} + C$$

$$6. \int e^x dx = e^x + C$$

### EXAMPLE PROBLEMS

**1. Evaluate**  $\int (x^2 + 4x + 1) dx$

**Solution :** Let  $I = \int (x^2 + 4x + 1) dx$

$$= \int x^2 dx + 4 \int x dx + \int 1 dx$$

$$= \frac{x^3}{3} + 4 \cdot \frac{x^2}{2} + x + C$$

$$= \frac{x^3}{3} + 2x^2 + x + C$$

**2. Evaluate**  $\int \left( 3e^x + \frac{4}{x} + 6 \right) dx$

**Solution :** Let  $I = \int \left( 3e^x + \frac{4}{x} + 6 \right) dx$

$$= \int 3e^x dx + \int 4 \cdot \frac{1}{x} dx + \int 6 \cdot dx$$

$$= 3 \int e^x dx + 4 \int \frac{1}{x} dx + \int 6 \cdot dx$$

$$= 3e^x + 4 \cdot \log x + 6x + C$$

**3. Evaluate**  $\int e^{5x+8} dx$

**Solution :** Let  $I = \int e^{5x+8} dx = \frac{e^{5x+8}}{5} + C$

**4. Evaluate**  $\int (1 + \sin 2x) dx$

**Solution :** Let  $I = \int (1 + \sin 2x) dx$

$$= \int 1 dx + \int \sin 2x dx$$

$$= x - \frac{\cos 2x}{2} + C$$

**5. Evaluate**  $\int \frac{\cos x}{\sin^2 x} dx$

**Solution :** Let  $I = \int \frac{\cos x}{\sin^2 x} dx = \int \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} dx$

$$= \int \cot x \cdot \operatorname{cosec} x dx = -\operatorname{cosec} x + C$$

**6. Evaluate**  $\int \frac{2}{\cos^2 x} dx$

**Solution :** Let  $I = \int \frac{2}{\cos^2 x} dx = 2 \int \frac{1}{\cos^2 x} dx$

$$= 2 \int \sec^2 x dx = 2 \tan x + C$$

**7. Integrate**  $\int (x-1)(x+1) dx$

**Solution :** Let  $I = \int (x-1)(x+1) dx$

$$I = \int (x^2 - 1) dx$$

$$| (a+b)(a-b) = a^2 - b^2$$

$$= \int x^2 dx - \int 1 dx = \frac{x^3}{3} - x + C$$

**8. Evaluate**  $\int \frac{1}{1 + \cos 2x} dx$

**Solution :** Let  $I = \int \frac{1}{1 + \cos 2x} dx = \int \frac{1}{2 \cos^2 x} dx$

$$= \frac{1}{2} \int \sec^2 x dx = \frac{1}{2} \tan x + C$$

**9. Evaluate**  $\int \frac{1}{1 - \cos 2x} dx$

**Solution :** Let  $I = \int \frac{1}{1 - \cos 2x} dx = \int \frac{1}{2 \sin^2 x} dx$

$$= \frac{1}{2} \int \operatorname{cosec}^2 x dx$$

$$= \frac{1}{2} (-\cot x) + C = -\frac{1}{2} \cot x + C$$